MIT Art Design and Technology University's

MIT School of Computing, Pune

Department of Computer Science and Engineering

Case Study

Subject – Environmental Sciences

Class - SY (SEM-II),

Division-SY-21

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CASE STUDY

Title: [Case Study on Kolhapur Floods]

1. Introduction

Background / Motivation:

Floods are among the most common and destructive natural disasters in India. Maharashtra, particularly the western districts like Kolhapur and Sangli, witnessed unprecedented floods in August 2019. Excessive rainfall in the Western Ghats, combined with continuous water discharge from dams such as Radhanagari, Koyna, and Warna, led to massive inundation across Kolhapur. Panchganga River and its tributaries swelled beyond their capacity, submerging towns, villages, farmlands, and highways.

Kolhapur is an agriculturally rich district, famous for sugarcane cultivation, jaggery production, and leather industries. The floods not only disrupted human life but also devastated agriculture, livestock, small businesses, and the overall economy of the district. Understanding the flood event in detail is crucial to design preventive and adaptive measures for future resilience.

Objectives:

- To study the meteorological and hydrological causes of the 2019 Kolhapur floods.
- To analyse the socio-economic and environmental impacts.
- To assess the flood response and relief operations carried out.
- To identify the lessons learned and suggest improvements for future disaster management.

2. Problem Statement

Description:

In August 2019, continuous rainfall caused rivers and dams in Kolhapur to overflow. Panchganga River crossed the danger mark multiple times, leading to waterlogging in major towns and villages. National Highway 4 (Mumbai–Bengaluru) was blocked for days, cutting off road transport. Over 200,000 people were evacuated, and more than 50 people lost their lives in Kolhapur and Sangli combined.

Scope:

This case study focuses specifically on Kolhapur district, covering the human, agricultural, and infrastructural impacts. While other regions like Sangli were also affected, the emphasis is on Kolhapur's unique geographical and economic vulnerability to floods.

3. Solution Overview

Approach:

• Early Warning & Evacuation – The district administration issued alerts through media and mobile messages. Local volunteers and police assisted in moving residents to higher ground.

• Rescue Operations – Teams from the National Disaster Response Force (NDRF), Army, Navy, and Coast Guard were deployed. Boats and helicopters were used for rescue.

•Relief & Rehabilitation – Relief camps were set up in schools and community halls. Food, water, medical aid, and clothes were distributed. The state government and NGOs provided immediate relief, followed by financial assistance for crop and property damage.

Technology Stack:

• Indian **Meteorological Department (IMD):** Rainfall forecasts and alerts.

• GIS **Mapping:** Flood-prone zones identified.

• Mobile **Communication:** WhatsApp and SMS used for quick alerts.

• Dam **Monitoring Systems:** Controlled release of water from Radhanagari and Koyna dams.

4. Implementation Details

System Architecture:

• Input**:** Rainfall data, river water levels, dam storage reports.

• Processing**:** IMD predictions, GIS flood mapping, disaster management committee discussions.

• Output**:** Evacuation orders, relief camp setup, coordinated rescue plans.

Key Components:

• Evacuation **Centres:** Around 200 relief camps were established.

• Rescue **Forces:** NDRF deployed over 35 teams in Kolhapur and Sangli.

• Medical **Response:** Special medical teams were mobilized to prevent cholera, malaria, and dengue outbreaks.

• Volunteer **Support:** Local NGOs like Bharat Sevashram Sangha, Red Cross, and community groups assisted in relief

Challenges Faced:

- **Sudden Water Rise:** Within hours, villages were submerged, leaving little time for preparation.
- **Transport Blockade:** NH-4 and rail routes were shut, delaying relief supplies.
- **Communication Gaps:** Dam authorities were criticized for lack of timely coordination with local administration.
- **Economic Burden:** Compensation was delayed for many farmers and small traders.

5. Results

Performance Metrics:

- **Evacuation:** Over 2.5 lakh people shifted to safer places.
- **Casualties:** More than 50 deaths, along with livestock loss in thousands.
- **Agriculture Loss:** Over 4 lakh hectares of sugarcane and paddy fields destroyed.
- **Infrastructure Damage:** Roads, bridges, schools, and power lines severely affected.

Comparison:

• Compared to floods in 2005, the 2019 event was more severe in terms of scale and economic loss.

• Use of technology like GIS and social media was better in 2019, but dam water release mismanagement remained a critical issue.

User Feedback:
Positive: Timely rescue operations saved thousands of lives.

6. Conclusion

Achievements:

- Successful large-scale evacuation with minimal loss of life considering the magnitude.
- Effective coordination between government, military, and NGOs.
- Better awareness among citizens about flood risks

Lessons Learned:

- Flood preparedness needs to include long-term planning, not just emergency response.
- Dam water management policies must be revised with inter-district coordination.
- Rural infrastructure (roads, drainage, embankments) must be flood-resilient.

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7. Future Enhancements

Proposed Improvements:

- Installation of **real-time flood monitoring systems** in rivers.
- Community-based disaster risk reduction programs.
- Strengthening of embankments and stormwater drainage systems.
- Establishing **flood insurance schemes** for farmers and small businesses.

Expansions:

- Integrating smart-city flood control systems in Kolhapur.
- Expanding regional cooperation between Maharashtra, Karnataka, and Goa for dam water management.

8. References

1. Government of Maharashtra – Disaster Management Report (2019).
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4. IITM Pune – Research papers on Western Ghats rainfall patterns.
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